Japan Sends a Spacecraft to Collect Pieces of a Mars Moon

From: Popular Science Mined: 2026-10-20 5 min read

Japan's space agency, JAXA, planned to launch a special mission to Phobos, one of the two moons of Mars, on Monday, and anyone could watch it live online.

The mission is called MMX, which stands for Martian Moons eXploration. Its goal is to collect **samples** from Phobos, the larger of Mars' two moons, and bring them back to Earth so scientists can study them. If the mission succeeds, the **samples** could help solve a long-standing mystery: how the moons of Mars formed.

An American **astronomer** named Asaph Hall discovered Phobos and its smaller sibling, Deimos, in 1877. Their names are pretty dramatic. Phobos comes from an ancient Greek word meaning "fear," and Deimos comes from a Greek word meaning "dread."

Even though scientists have paid a lot of attention to Mars, they still do not know exactly where its moons came from. One idea is that they were once **asteroids**. Another is that they formed from leftover material from the early days of the **solar system**, about 4.5 billion years ago.

Although MMX is a Japanese mission, NASA helped by providing **instruments** and technology. Nicky Fox, a NASA science leader, said the **samples** "will bring forth **paradigm-shifting** science on the Mars system for generations to come," especially as NASA prepares to send the first humans to Mars.

The spacecraft was set to launch on an H3 rocket from Japan's Tanegashima Space Center at 3:41 p.m. Eastern time on Monday, October 19. That was 4:41 a.m. on October 20 in Japan. The launch could be watched on JAXA's YouTube page.

The trip is a long one. MMX is expected to reach Phobos in August 2027 and study its surface for at least 16 months before collecting **samples**. The **samples** will not come back to Earth until at least 2031.

Word Treasure

samples -- Small pieces of something collected so scientists can study it closely.

astronomer -- A scientist who studies stars, planets, moons and other objects in space.

asteroids -- Rocky objects that orbit the Sun and are smaller than planets.

solar system -- The Sun and everything that travels around it, including planets, moons and asteroids.

paradigm-shifting -- So important that it changes the way experts think about a subject.

instruments -- Tools and devices used to measure or study things, like cameras or sensors on a spacecraft.

Background you need

Mars has two small moons, Phobos and Deimos, which are much smaller and more oddly shaped than Earth's Moon.

A sample-return mission sends a spacecraft to collect material from another world and carry it back so scientists can study it with powerful lab tools on Earth.

Space agencies often work together, sharing instruments and technology so that one mission can do more science.

Because space is so vast, trips to Mars and back take years, so scientists plan these missions far ahead.

Break it down

LEAD: Japan's JAXA planned to launch the MMX mission to the Mars moon Phobos on Monday

KEY EVENT: MMX aims to collect samples from Phobos and return them to Earth

+- WHO: JAXA, with instruments and technology from NASA
+- WHERE: Launch from Tanegashima Space Center on an H3 rocket
+- WHEN: 3:41 p.m. Eastern time on October 19 (October 20 in Japan)

BACKGROUND: Asaph Hall discovered Phobos and Deimos in 1877; their origin is still unknown

TIMELINE: Arrive August 2027, survey at least 16 months, samples back by at least 2031

OPEN QUESTION: Were Mars' moons once asteroids, or leftovers from the early solar system?

Why it matters

Bringing back real pieces of a Mars moon could solve a long-standing mystery about how the moons formed and help prepare for future human trips to Mars.

Test what you remember

1. What is the main goal of the MMX mission?

- (A) To collect samples from Phobos and bring them to Earth
- (B) To build a base on the Moon
- (C) To land astronauts on Mars
- (D) To take photos of Jupiter

2. What does the name Phobos mean in ancient Greek?

- (A) Dread
- (B) Light
- (C) Speed
- (D) Fear

3. Who discovered Phobos and Deimos in 1877?

- (A) Carson Bruns
- (B) Nicky Fox
- (C) Asaph Hall
- (D) Lord Kelvin

4. How did NASA help with this Japanese mission?

- (A) It did not help
- (B) It provided instruments and technology
- (C) It built the rocket
- (D) It chose the mission's name

5. When is MMX expected to reach Phobos?

- (A) 2050
- (B) January 2026
- (C) August 2027
- (D) Next week

6. What is one idea about how Mars' moons formed?

- (A) They were built by people
- (B) They are made of ice cream
- (C) They broke off from Earth
- (D) They were once asteroids

Answer key (try the quiz first!): 1: A 2: D 3: C 4: B 5: C 6: D

Questions to think about

1. Nicky Fox, NASA: The samples will bring paradigm-shifting science on the Mars system for generations, especially as NASA prepares to send humans to Mars.

MY THOUGHTS (at least 20 words):